A cross-cohort single-cell reference atlas of H3K27-altered diffuse midline glioma

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Abstract

Diffuse midline glioma, H3 K27-altered (DMG), is a lethal brain tumour, mostly of children, whose single-cell data are spread across small cohorts that differ in assay, anatomical site and clinical context. We integrated six public human DMG datasets into a reference of 311,618 cells and nuclei from 57 samples and 52 patients. Copy-number-based malignant calls agreed with the original annotations in a median 0.898 of cells per sample, and lineage labels transferred to held-out patients with 0.975 accuracy. Under a replication rule specified before analysis, 2 of 15 malignant expression metaprograms, cycling and hypoxia, recurred across three studies and in both cells and nuclei; every lineage-like programme was detected only in nuclei, but only six whole-cell samples were available. In 21 spatial transcriptomics sections from 12 patients, a pre-specified association between mesenchymal-like tumour cells and myeloid cells was supported and an oligodendroglial association was not. Refitting the reference without each study showed that malignant and myeloid labels transfer to unseen studies, whereas rare normal lineages are frequently misclassified as malignant (balanced accuracy 0.40–0.99 per held-out study), and that cycling is the only metaprogram recovered in every scenario. The atlas defines which DMG cell states are reproducible across cohorts and provides openly available data, reference models and code for mapping new samples.

Keywords: diffuse midline glioma; H3 K27 alteration; single-cell RNA sequencing; single-nucleus RNA sequencing; spatial transcriptomics; data integration; reference atlas

Introduction

Diffuse midline glioma, H3 K27-altered, is a World Health Organization (WHO) grade 4 tumour of the pons, thalamus and spinal cord. Median survival is under one year, and no systemic therapy has improved it[1–3]. Most tumours carry a lysine-to-methionine substitution at position 27 of histone H3 (H3K27M)[4,5], which inhibits Polycomb repressive complex 2 (PRC2) and causes global loss of H3K27 trimethylation with focal retention and widespread H3K27 acetylation[6–9]. Transcriptional dependencies[10], electrical integration into neural circuits[11] and subclonal functional diversity[12] have been described, and cellular immunotherapy has shown activity[13]. The imipridone dordaviprone (ONC201) produced responses in recurrent H3K27M-mutant DMG[14] and received accelerated approval from the US Food and Drug Administration in 2025 as the first systemic therapy for this tumour, but durable remission remains rare.

Single-cell RNA sequencing showed that DMG is dominated by oligodendrocyte-precursor-like (OPC-like) cells, with smaller astrocyte-like (AC-like) and oligodendrocyte-like (OC-like) fractions[15], a model later refined by developmental anchoring[16,17] and age-stratified analysis[18]. This hierarchy resembles that of adult glioblastoma[19,20]. The tumour microenvironment is non-inflammatory and dominated by microglia[21,22], candidate immune checkpoints have been proposed[23], and spatial studies have begun to locate tumour states in tissue[24,25].

Three problems limit this literature. First, individual cohorts are small, so patient-level inference is rarely possible. Second, deposits differ in ways that are confounded with biology: pontine tumours are often sampled at autopsy and profiled as frozen nuclei[26], whereas thalamic tumours are more often resected and profiled as whole cells[27]. Third, tumour cell states are usually defined within a single study, so it is unclear which reflect the disease and which reflect the protocol[28]. Integrated reference atlases have addressed similar problems in other tissues and in adult glioblastoma[29,30], but, to our knowledge, no cross-cohort reference has been published for DMG.

Here we built such a reference and used it to test which findings hold across cohorts. We integrated six public datasets with variational autoencoders[31,32], benchmarked the integration[33] and projected five further cohorts onto it[34]. We make four contributions: (i) a harmonised 52-patient reference with calibrated malignant calls and reusable models; (ii) a replication test of malignant expression programmes against a rule specified before analysis; (iii) patient-level spatial tests of two pre-specified co-localisation hypotheses; and (iv) a leave-one-study-out evaluation that measures what the reference can and cannot be trusted to do on new data.

Results

An integrated reference of 52 patients

We selected six public studies as the reference and reserved five further cohorts for projection only (Supplementary Table 1). Of 420,381 cells in eligible samples, 356,406 (84.8%) passed per-assay quality control. After restriction to samples with unique molecular identifier (UMI) counts, the reference comprised **311,618 cells and nuclei from 57 samples and 52 patients**, measured over 18,882 genes shared by all studies (Table 1, Fig. 1a). The post-mortem series of Okada et al. contributed half of the cells (156,936 nuclei from 22 patients)[26], followed by the single-cell and multiome series of Jessa et al. (129,364 cells from 23 patients)[16] and the cohorts

of Collot et al.[23], Zang et al.[35] and DeSisto et al.[27]. Nuclei made up 88.4% of the reference; 42 samples were pontine and 15 thalamic. The H3 variant was reported for 32 of 57 samples (8 H3.1, 1 H3.2 and 23 H3.3).

We integrated the data with single-cell variational inference (scVI)[31], using sample as the batch variable and study, assay and mitochondrial fraction as covariates, and benchmarked three random seeds against Harmony[36] and uncorrected principal components with single-cell integration benchmarking (scIB) metrics[33]. A pre-specified score (0.4 × batch correction + 0.6 × biological conservation, min–max scaled across candidates) selected seed 1, which scored 0.739 (batch 0.537, biology 0.874), compared with 0.581 for Harmony and 0.126 for principal components (Fig. 1e, Supplementary Fig. 2). Lineage compartments separated clearly and studies intermixed within them (Fig. 1b–d). Mixing across assays remained incomplete: the integration local inverse Simpson's index (iLISI) for assay was 0.029 for the selected model and 0.134 for Harmony.

Table 1 | Reference cohort. Samples, patients and cells per contributing study after quality control, with assay, anatomical site and reported H3 variant (sample counts), and the fraction of cells called malignant by the consensus rule. Counts are cells for whole-cell and nuclei for single-nucleus and multiome assays. Patient counts per study sum to 56 because the two Jessa et al. series share four patients; the total gives distinct patients.

Study	Samples	Patients	Cells	Assay (cells)	Site (samples)	H3 variant (samples)	Malignant
Okada 2026	22	22	156,936	snRNA:156936	thalamus:6;pons:16	unknown:22	0.516
Jessa 2022 (sc/snRNA)	17	17	83,076	snRNA:54545;scRNA:28531	pons:13;thalamus:4	H3.1:6;H3.2:1;H3.3:10	0.566
Jessa 2022 (multiome)	10	10	46,288	multiome_RNA:46288	pons:9;thalamus:1	H3.1:2;H3.3:8	0.595
Collot 2026	5	4	17,562	snRNA:17562	pons:3;thalamus:2	H3.3:5	0.570
Zang 2024	1	1	6,546	scRNA:6546	pons:1	unknown:1	0.710
DeSisto 2024	2	2	1,210	scRNA:1210	thalamus:2	unknown:2	0.631
Total	57	52	311,618	18,882 genes shared by all studies			

Malignant calls agree with published labels

Malignant status was assigned by combining lineage-marker scores with an expression-based copy-number profile computed against normal cells of the same sample[19] (Methods). Across **324,676 author-annotated cells in 97 samples**, the copy-number call agreed with the authors' labels in a median **0.898** of cells per sample (interquartile range (IQR) 0.793–0.978; pooled agreement 0.884, precision 0.934, recall 0.909; Table 2). Agreement in the 22 samples used to tune thresholds (0.882) was similar to that in the 75 remaining samples (0.885). Calls supported by a recurrent arm-level event were more precise than calls based on the aggregate score alone (0.952 versus 0.862), so event support is released as a confidence field.

Gain of chromosome 1q was the most frequent event, followed by loss of 14q and 13q (Fig. 1f). We excluded 4q12 from the event set because, at 100-gene resolution, it measures OPC-lineage expression rather than copy number. Within the reference, the consensus of marker and copy-number evidence classified 170,940 cells as malignant (54.9%), 50,754 as normal (16.3%) and 89,924 as ambiguous (28.9%). A permissive variant that accepts event-supported copy-number calls classified a further 28,252 cells as malignant and is released as a sensitivity label.

Labels transfer to new patients and external cohorts

A semi-supervised classifier (scANVI)[32], evaluated on five patients withheld from classifier training, reached **0.975 accuracy** and 0.917 balanced accuracy on 23,827 cells. Recall ranged from

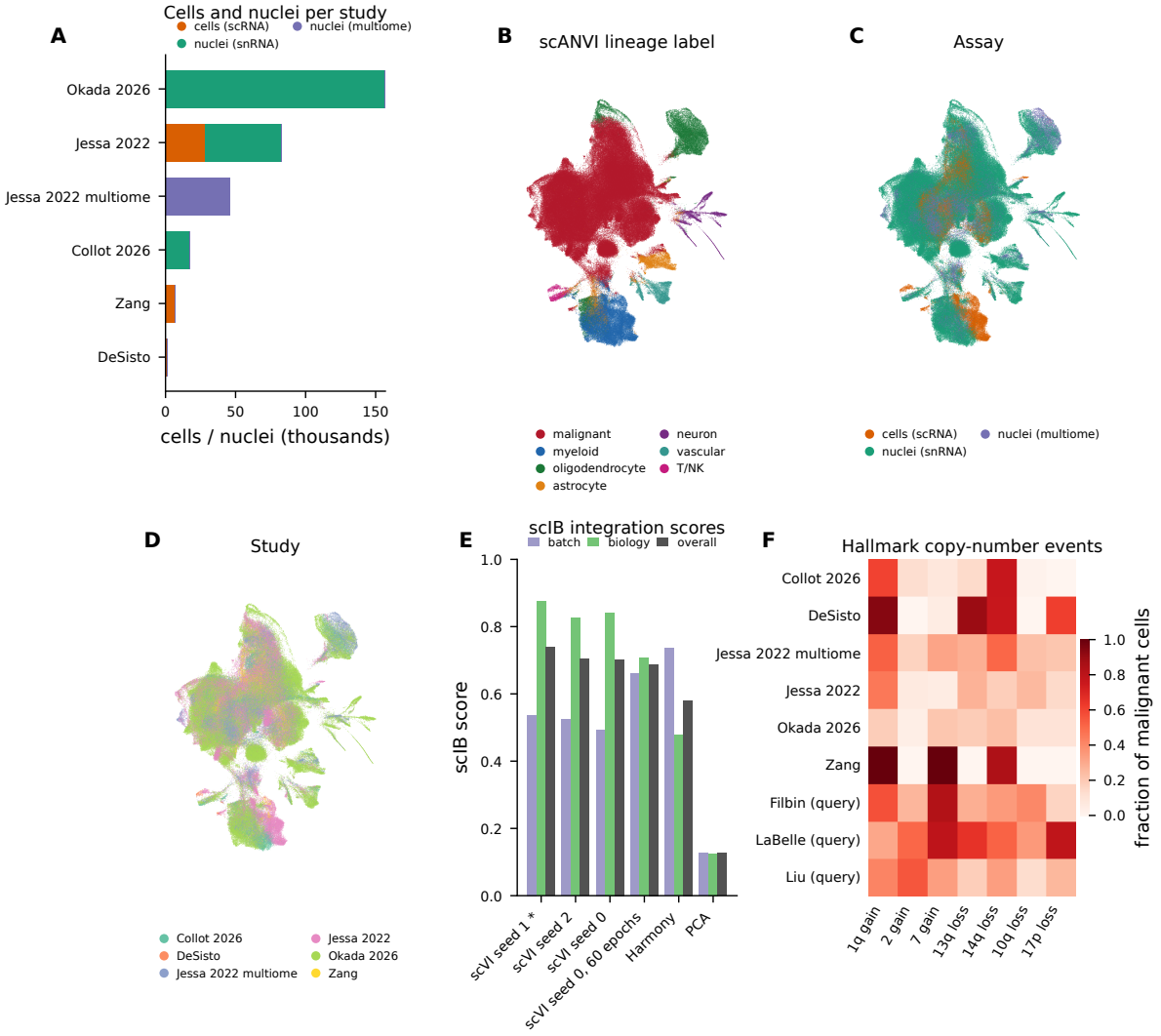


Fig. 1 | An integrated reference of H3 K27-altered diffuse midline glioma. a, Cells and nuclei per study, coloured by assay. **b–d**, UMAP of the scANVI latent space (311,618 cells, 57 samples, 52 patients) coloured by lineage (b), assay (c) and study (d). **e**, scIB metrics for three scVI seeds, a 60-epoch scVI sensitivity run, Harmony and uncorrected principal components; the asterisk marks the model selected by the pre-specified score. **f**, Fraction of copy-number-malignant cells carrying each recurrent arm-level event, by study, including the three projection-only tumour cohorts.

Table 2 | Agreement of malignant calls with published annotations. Tier-2 comparison of the copy-number call against each study's own annotation, for every annotated cell. Pooled values weight cells equally; the sample median (with interquartile range) and unweighted sample mean weight samples equally.

Study	Cells	Samples	Agreement	Precision	Recall	Sample median (IQR)	Sample mean
Jessa 2022 (sc/snRNA)	84,194	17	0.915	0.982	0.914	0.898 (0.835–0.964)	0.846
Jessa 2022 (multiome)	46,618	10	0.935	0.948	0.967	0.948 (0.897–0.988)	0.930
Collot 2026	17,750	5	0.907	0.896	0.975	0.960 (0.919–0.964)	0.929
Okada 2026	161,043	22	0.850	0.906	0.890	0.868 (0.763–0.895)	0.800
Liu 2022 (query)	8,705	42	0.799	0.975	0.782	0.945 (0.770–0.990)	0.813
LaBelle (query)	6,366	1	1.000	nan	nan	1.000 (1.000–1.000)	1.000
All	**324,676**	**97**	**0.884**	**0.934**	**0.909**	**0.898 (0.793–0.978)**	**0.836**

0.999 for malignant cells to 0.803 for astrocytes (Supplementary Fig. 2). Because the underlying scVI encoder had seen these patients, this test measures generalisation to new patients from known studies; generalisation to new studies is tested below.

We next projected five cohorts excluded from training through the frozen encoder without fine-tuning (Fig. 2). Most cells of the two DMG cohorts deposited as transcripts per million (TPM) were classified as malignant (median 90.3% per sample for Filbin et al.[15] and 83.6% for the 42 samples from 41 patients of Liu et al.[18]), and 95.1% of the cells that Liu et al. annotated as malignant were classified as malignant. In the immune-focused cohort of LaBelle et al.[37], the median per-sample malignant fraction was 49.9% and the pooled fraction 11.1%. Adult pons nuclei[38] were classified as malignant in a median of 1.2% of cells, whereas fetal hindbrain cells[39] were classified as malignant in a median of 91.3%. The reference contains no normal progenitor class, so fetal radial glia, glioblasts and neuroblasts were assigned to the nearest available class. Their median distance to the 15 nearest reference cells was 2.60, compared with 1.57 among reference cells, but the TPM cohorts showed similarly high mapping uncertainty, so neighbour distance alone does not identify this failure (Supplementary Fig. 3).

Two of 15 metaprograms replicate

Consensus non-negative matrix factorisation (cNMF)[40] of 167,012 malignant cells from 50 samples yielded per-sample expression programmes, which we clustered into 15 metaprograms, that is, programmes recurring across samples (Table 3, Fig. 3a, Supplementary Fig. 4). Twelve had a lineage-like identity and three were technical (ribosomal, mitochondrial and dissociation stress). Under the pre-specified rule, a metaprogram replicates if it recurs in at least three studies and in both cells and nuclei. **Two metaprograms replicated**: cycling (MP1, 34 samples) and hypoxia (MP4, 15 samples). Both also replicated under a stricter rule requiring at least two samples per assay class. This count was stable across dendrogram cuts from 0.84 to 0.90 (Supplementary Fig. 4c).

Every lineage-like metaprogram was detected only in nuclei, and the only programme restricted to whole cells reflected dissociation stress. This result is limited by sample size, because only six whole-cell samples from three publications contributed to programme discovery. Masking 6,013 genes that differ between paired cells and nuclei yielded 12 metaprograms, 3 of which replicated: cycling, hypoxia and an OPC-like metaprogram derived mainly from MP3. The OPC-like metaprogram included a single whole-cell sample and did not replicate under a stricter rule requiring at least two samples per assay class, under which 2 metaprograms replicated. Random masks of the same size left a median of 2.5 metaprograms, 1 of which replicated; because random masks removed far more programme structure, this control does not isolate the effect of assay-biased genes (Fig. 3b, Supplementary Fig. 5). The ciliated metaprogram (MP9; *ADGB*, *CFAP43*, *DNAH9*) consists of motile-cilia genes that are also expressed by normal ependymal cells, which the reference lacks; it may therefore reflect ependymal contamination rather than a tumour state.

Microenvironment and composition

Subclustering resolved 21 microenvironment populations, including homeostatic microglia, three tumour-associated macrophage states, $CD4^+$ and $CD8^+$ T cells, natural killer cells, five vascular

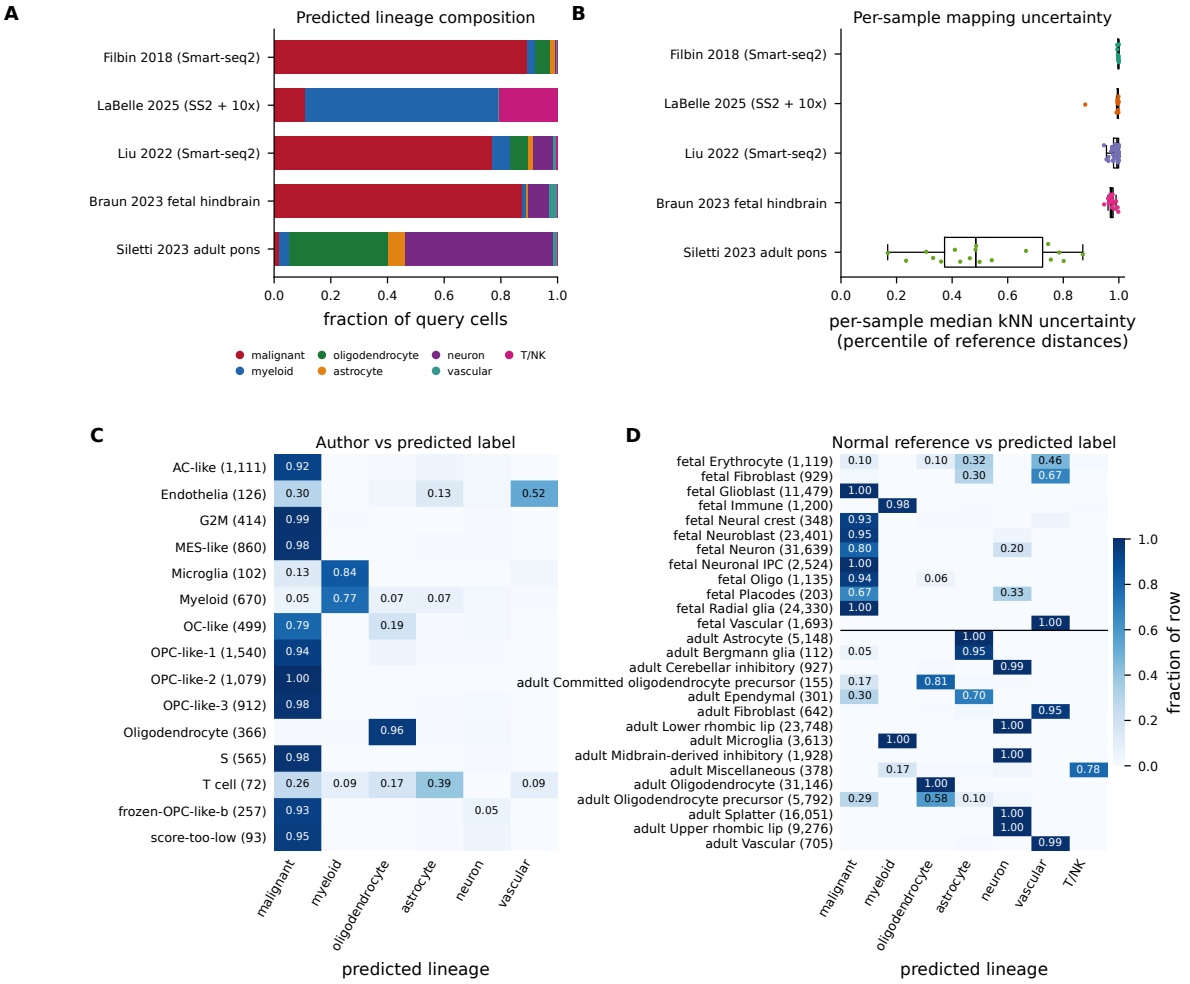


Fig. 2 | Zero-shot projection of five external cohorts. **a**, Predicted lineage composition of each query cohort. **b**, Per-sample median neighbour-based uncertainty, expressed as a percentile of the reference's own 15-nearest-neighbour distances. **c**, Author-annotated state against predicted lineage for Liu et al. (row-normalised). **d**, Fetal (Braun et al.) and adult (Siletti et al.) reference classes against predicted lineage.

Table 3 | Consensus metaprograms. Metaprograms at the primary dendrogram cut, their assigned lineage family or technical class, the number of samples in which a contributing programme was found, whether they satisfy the pre-specified replication rule (at least three studies and both assay classes), and their top genes.

Metaprogram	Family	Samples	Replicated	Top genes
MP1	cycling	34	yes	*ASPM, TOP2A, CENPF, MKI67, KIF14, TPX2*
MP2	*ribosomal*	18	no	*RPL32, RPS18, RPS8, RPS27, EEF1A1, RPL8*
MP3	OPC-like	18	no	*FGF12, DSCAM, TNR, CA10, DCC, CADM2*
MP4	hypoxia	15	yes	*SLC6A6, NDRG1, AKAP12, VEGFA, HILPDA, IGFBP5*
MP5	OC-like	12	no	*TNS3, BCAS1, OPCML, GPR17, PKP4, FGF12*
MP6	OPC-like	12	no	*LHFPL3, SEZ6L, LRRC4C, KCNIP4, MMP16, SEMA5A*
MP7	AC-like	8	no	*DAAM2, NKAIN3, MGAT4C, EEPD1, DCLK1, SLC4A4*
MP8	MES-like	7	no	*CD44, NAMPT, HRH1, SAMD4A, TGFB2, TPST1*
MP9	ciliated (MP9)	6	no	*ADGB, CFAP43, DNAH9, ZBBX, MAP3K19, CFAP299*
MP10	AC-like	6	no	*SLC4A4, EEPD1, FAT3, PTPRT, ADGRV1, SLC1A3*
MP11	cycling	5	no	*BRIP1, DTL, CENPP, CENPK, RFC3, MMS22L*
MP12	*mitochondrial*	5	no	*MT-CO2, MT-ND4, MT-CO3, MT-ND3, MT-CYB, MT-ATP6*
MP13	*IEG / dissociation stress*	5	no	*IER2, FOS, JUNB, EGR1, JUN, FOSB*
MP14	AC-like	5	no	*DCLK1, EEPD1, SORBS1, AQP4, DAAM2, RGMA*
MP15	AC-like	5	no	*EEPD1, CD44, SPARCL1, SORBS1, SLC4A4, TNC*

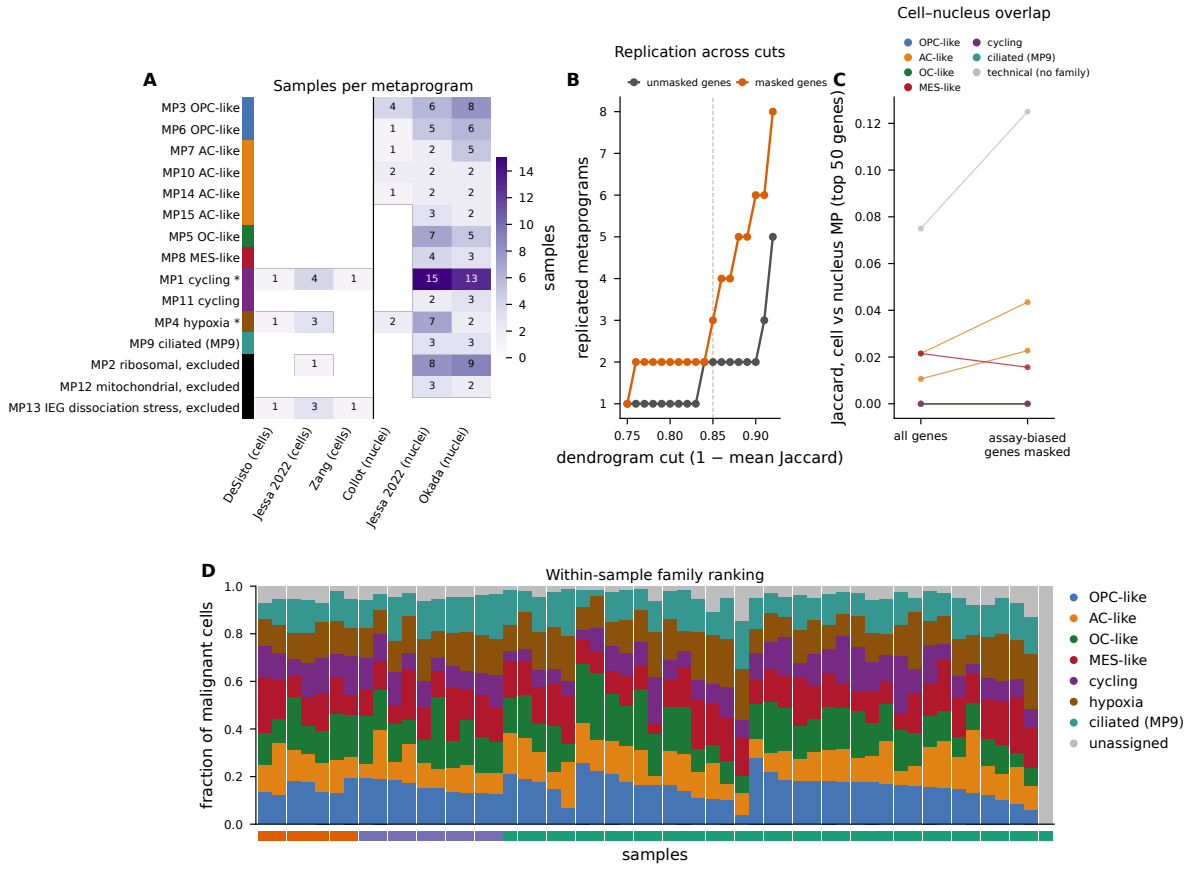


Fig. 3 | Metaprograms and their assay dependence. a, Metaprogram occurrence by study and assay; replicated metaprograms are marked and technical programmes are labelled. **b**, Replicated metaprograms across dendrogram cuts, with and without masking the 6,013 assay-biased genes; the dashed line marks the primary cut. **c**, Jaccard similarity of matched cell- and nucleus-derived metaprograms before and after masking. **d**, Per-sample distribution of malignant cells across metaprogram families. Families are assigned from scores standardised within each sample, so this panel shows the relative ranking of programmes within samples, not absolute family abundance.

populations and five oligodendroglial states (Fig. 4a, Supplementary Fig. 6). This composition is consistent with a myeloid-dominated, lymphocyte-poor microenvironment[21,22]. Several populations were derived almost entirely from nuclei (Fig. 4b).

Across samples, neurons were more abundant in H3.1 than in H3.3 tumours (change in logit proportion 2.60, $q = 0.004$; site-adjusted $q = 0.017$; pons only $q = 0.050$). All H3.1 samples, however, were pontine nuclei from a single publication, so this difference cannot be separated from study. Astrocytes (change in logit proportion 1.76, $q = 0.038$) and myeloid cells (1.85, $q = 0.038$) were also more abundant in H3.1 tumours, but neither remained significant after adjustment for site ($q = 0.167$). Among microenvironment subclusters, one mature-oligodendrocyte subcluster was more abundant in H3.1 tumours ($q = 0.011$; site-adjusted $q = 0.009$), and several oligodendroglial and vascular subclusters differed between the 6 adult and 46 paediatric patients ($q < 0.05$). All composition results are provided in Supplementary Data 6. The largest compositional difference was technical: neurons were more abundant in nuclei than in whole cells (change in logit proportion 3.40, $q < 10^{-9}$; Fig. 4c).

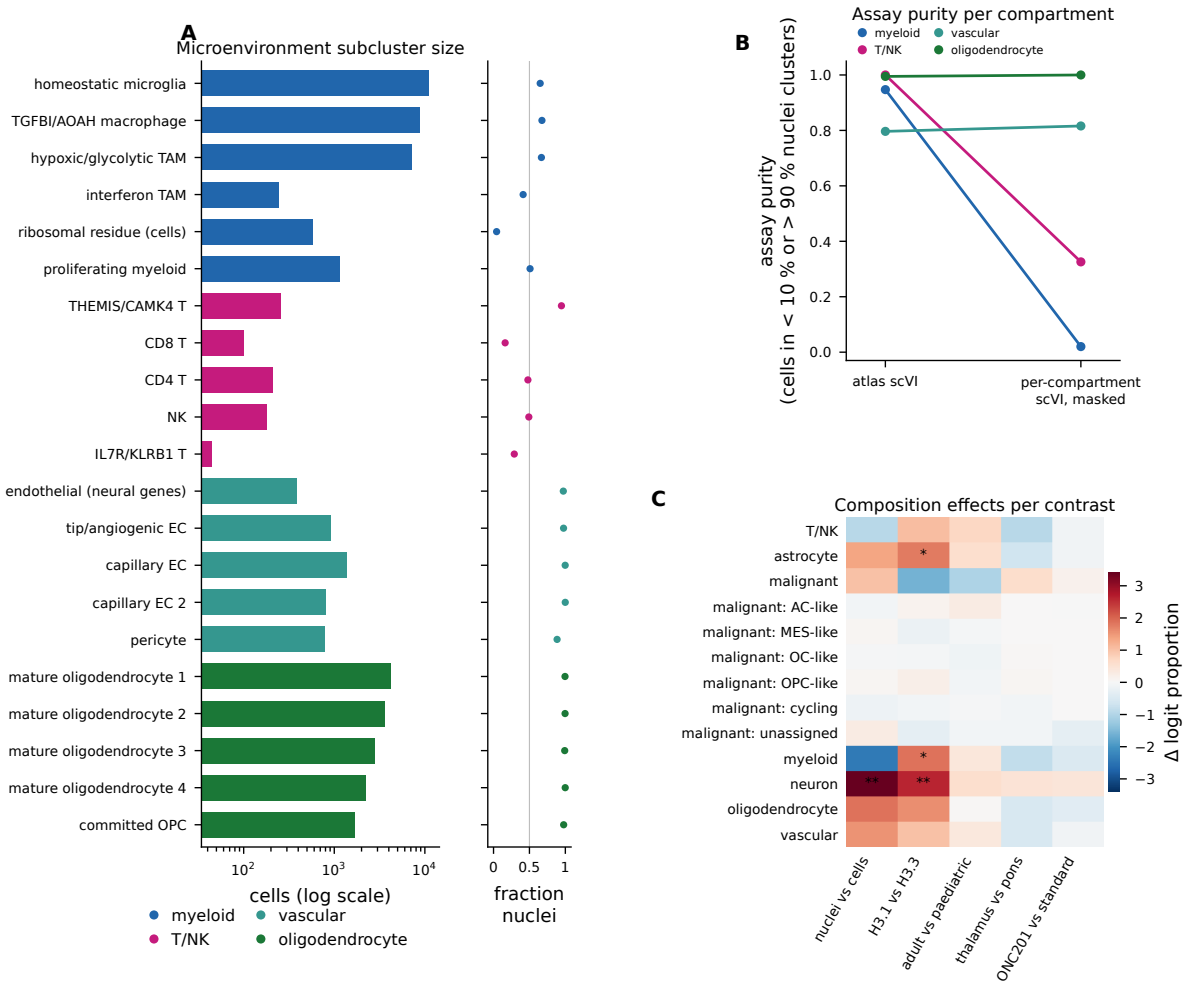


Fig. 4 | Microenvironment populations and sample-level composition. **a**, Microenvironment subclusters: cell count (log scale) and fraction derived from nuclei. **b**, Assay purity of microenvironment clusters in the atlas embedding and in a per-compartment model. **c**, Composition contrasts as change in logit proportion; asterisks mark Benjamini–Hochberg significance[41].

Spatial co-localisation of malignant states

We deconvolved 27 Visium spatial transcriptomics sections from four cohorts with cell2location[42], using a 13-class reference, and tested co-localisation on spot proportions with patients as the unit of inference (21 DMG sections from 12 patients; Table 4, Fig. 5). AC-like tumour cells co-localised with normal astrocytes ($\rho = 0.393$, $q < 0.001$) and oligodendrocytes ($\rho = 0.308$, $q = 0.005$). Of the two pre-specified hypotheses, mesenchymal-like (MES-like) tumour cells co-localised with myeloid cells ($\rho = 0.238$, 95% confidence interval (CI) 0.071–0.392, $q = 0.034$), whereas OPC-like tumour cells and oligodendrocytes were not associated ($\rho = 0.178$, 95% CI −0.028 to 0.370, $q = 0.210$). These associations were not consistent across sections: the correlation was positive in 13, 14 and 11 of 21 sections for the three pairs, the mean per-section correlations (0.130, 0.171 and 0.079) were smaller than the pooled estimates, and heterogeneity between patients was high for every pair (I^2 0.952–0.997). With 12 patients, the random-effects confidence intervals are likely to be too narrow, and these associations should be regarded as tentative (Supplementary Fig. 7). With centred log-ratio-transformed proportions, the MES-like–myeloid ($\rho = 0.238$, $q = 0.012$) and AC-like–astrocyte ($\rho = 0.230$, $q = 0.001$) associations persisted, the AC-like–oligodendrocyte association weakened ($\rho = 0.129$, $q = 0.048$), and the OPC-like–oligodendrocyte association was absent ($\rho = 0.007$, $q = 0.937$).

In ten sections from different levels of one brainstem autopsy (medulla, pons, midbrain and spinal cord)[43], tumour fraction increased with the AC-like and OPC-like shares and decreased with the normal glial, vascular and neuronal shares; with a single patient, this observation is descriptive. Five DMG and five glioblastoma sections from one study[24] differed in tumour fraction (0.528 versus 0.460) and cycling share (0.112 versus 0.214), but neither difference was significant after correction ($q = 0.667$).

Table 4 | Spatial co-localisation of malignant states with microenvironment lineages. Spot-level Spearman correlations of cell2location proportions, Fisher-z averaged within patient and pooled by random effects over 12 DMG patients and 21 sections, Benjamini–Hochberg corrected across 42 state-by-microenvironment pairs. The ten strongest pairs are shown; I^2 quantifies between-patient heterogeneity.

Malignant state	Lineage	ρ	95% CI	q	Sections > 0	I^2
AC-like	astrocyte	0.393	0.234 to 0.531	9.38e-05	13/21	0.993
ciliated (MP9)	neuron	0.230	0.131 to 0.324	9.38e-05	16/21	0.978
ciliated (MP9)	astrocyte	0.369	0.216 to 0.505	9.38e-05	13/21	0.992
AC-like	oligodendrocyte	0.308	0.138 to 0.459	0.00512	11/21	0.993
MES-like	vascular	0.253	0.078 to 0.412	0.0339	13/21	0.993
MES-like	myeloid	0.238	0.071 to 0.392	0.0339	14/21	0.992
ciliated (MP9)	oligodendrocyte	0.235	0.073 to 0.385	0.0339	13/21	0.992
OC-like	vascular	-0.214	-0.367 to -0.051	0.0511	6/21	0.993
ciliated (MP9)	T/NK	0.188	0.044 to 0.324	0.0511	14/21	0.990
hypoxia	T/NK	0.082	0.017 to 0.147	0.0589	12/21	0.952

Robustness to removing one study

Two publications contributed 92% of the cells, so we repeated the central analyses with each of the five publications removed in turn; the two Jessa et al. series share four patients and were treated as one publication.

For label transfer, scVI and scANVI were retrained without the held-out publication, which

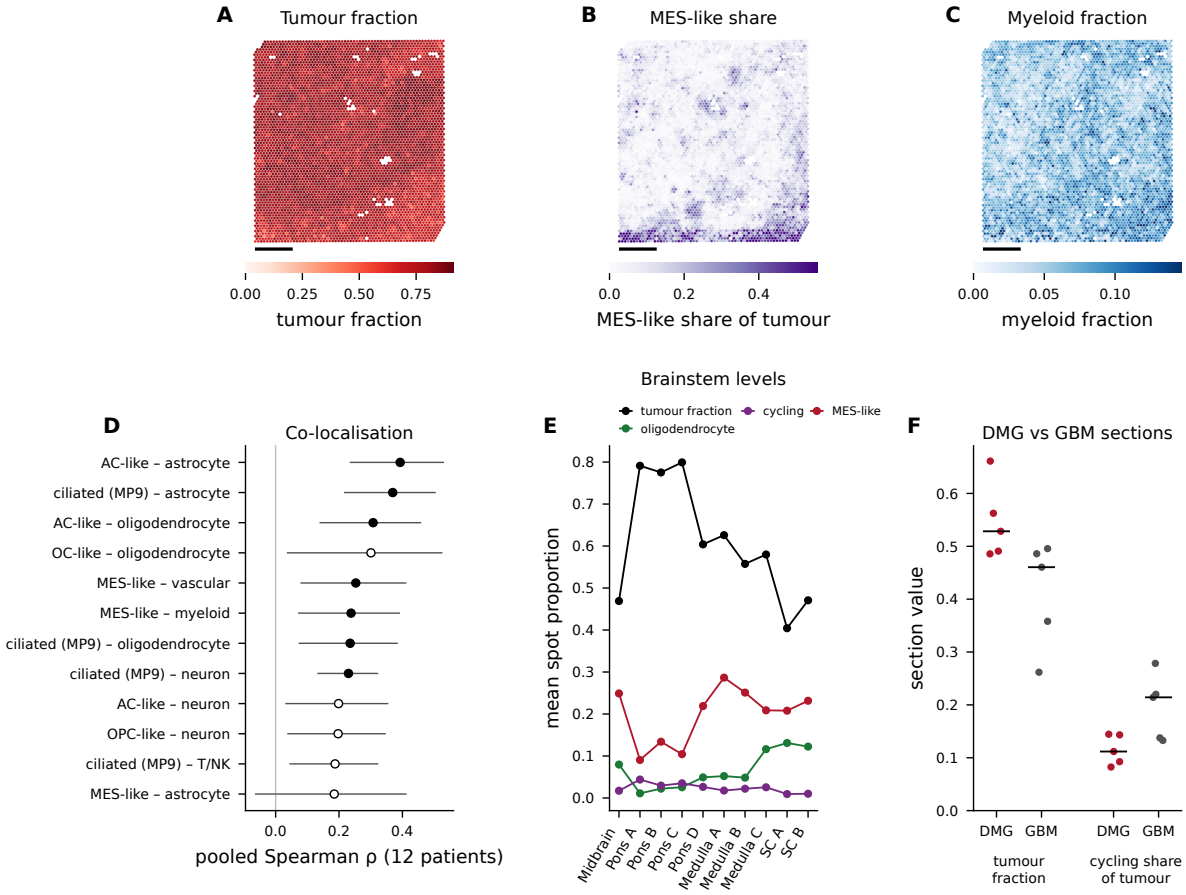


Fig. 5 | Spatial organisation of malignant states. a–c, A representative section (scale bar, 1 mm) showing tumour fraction, the mesenchymal share of tumour and myeloid fraction. **d**, Pooled spot-level Spearman correlations between malignant states and microenvironment lineages (12 patients, 21 sections); filled points are significant after Benjamini–Hochberg correction across 42 pairs. **e**, Mean spot proportions in ten sections from different brainstem levels of one autopsy. **f**, Tumour fraction and cycling share in five DMG and five glioblastoma sections from one study.

was then mapped by architectural surgery[34] with its labels hidden. Accuracy, computed on cells with a malignant or normal consensus label, remained between 0.880 and 0.992, but balanced accuracy ranged from 0.402 to 0.988 (0.506 when all folds were pooled by cell, compared with 0.917 for held-out patients; Fig. 6a,b). Malignant (recall 0.86–1.00) and myeloid (recall 0.98–1.00) labels transferred to every study, whereas astrocyte recall ranged from 0.00 to 0.56 and T and natural killer cells were mostly classified as myeloid. Errors favoured the malignant class: when the post-mortem series was held out, 98% of its neurons (3,838 of 3,898), 63% of its astrocytes and 51% of its vascular cells were classified as malignant (Fig. 6c). In this fold, the errors were made with high confidence: accuracy among predictions with probability of at least 0.9 was 0.912. Because this series contributes half of the reference, holding it out also halves the training data, so the drop in performance reflects both study shift and reduced training size.

For metaprograms, we repeated the clustering without each publication, in 20 publication-balanced pools of at most five samples per publication, and in 20 random pools of the same size. Cycling (MP1) was the only metaprogram recovered in every scenario and replicated in all 20 balanced pools; three further metaprograms (ribosomal MP2, OC-like MP5 and OPC-like MP6) were recovered whenever a single publication was removed but not in most smaller pools. Hypoxia (MP4) was lost when the Jessa et al. series was removed and replicated in only 4 of 20 balanced pools. The MES-like and ciliated metaprograms, and four others, were recovered only when both large series were present (Fig. 6d).

For composition, we relabelled the ambiguous cells (28.9% of the reference) in three ways: consensus labels only, ambiguous as a separate category, and confident scANVI labels only. All four significant effects remained significant and retained their sign under every labelling (Fig. 6e).

Discussion

We present a reference of H3 K27-altered DMG built from 52 patients, with harmonised labels, calibrated malignant calls and reference models whose performance on new patients and on new studies has been measured. Because individual DMG cohorts contain few patients, this reference allows patient-level questions that no single public dataset can address.

Most of our findings define the limits of what public DMG data can establish. Only cycling and hypoxia replicated under a rule specified in advance, and of these only cycling was recovered in every robustness scenario. Lineage-like programmes were detected only in nuclei, but with six whole-cell samples available for programme discovery we cannot distinguish an effect of assay from limited statistical power; matched whole-cell and nucleus data from the same tumours are needed. This mirrors pan-cancer evidence that recurrent programmes are identified reliably only across many tumours[28], and adds that recurrence across cohorts is insufficient when cohorts differ systematically in sample preparation. Similarly, the H3.1–H3.3 difference in neuron abundance and the differences between cells and nuclei are confounded with study by design.

The leave-one-study-out analysis has a practical implication for tumour reference atlases. Accuracy on held-out patients, a commonly reported measure, overstated performance on a new study (balanced accuracy 0.917 versus 0.506). A reference composed mainly of tumour tissue transferred malignant and myeloid identities reliably but classified unfamiliar normal cells as malignant, with high confidence. We therefore recommend leave-one-study-out evaluation for

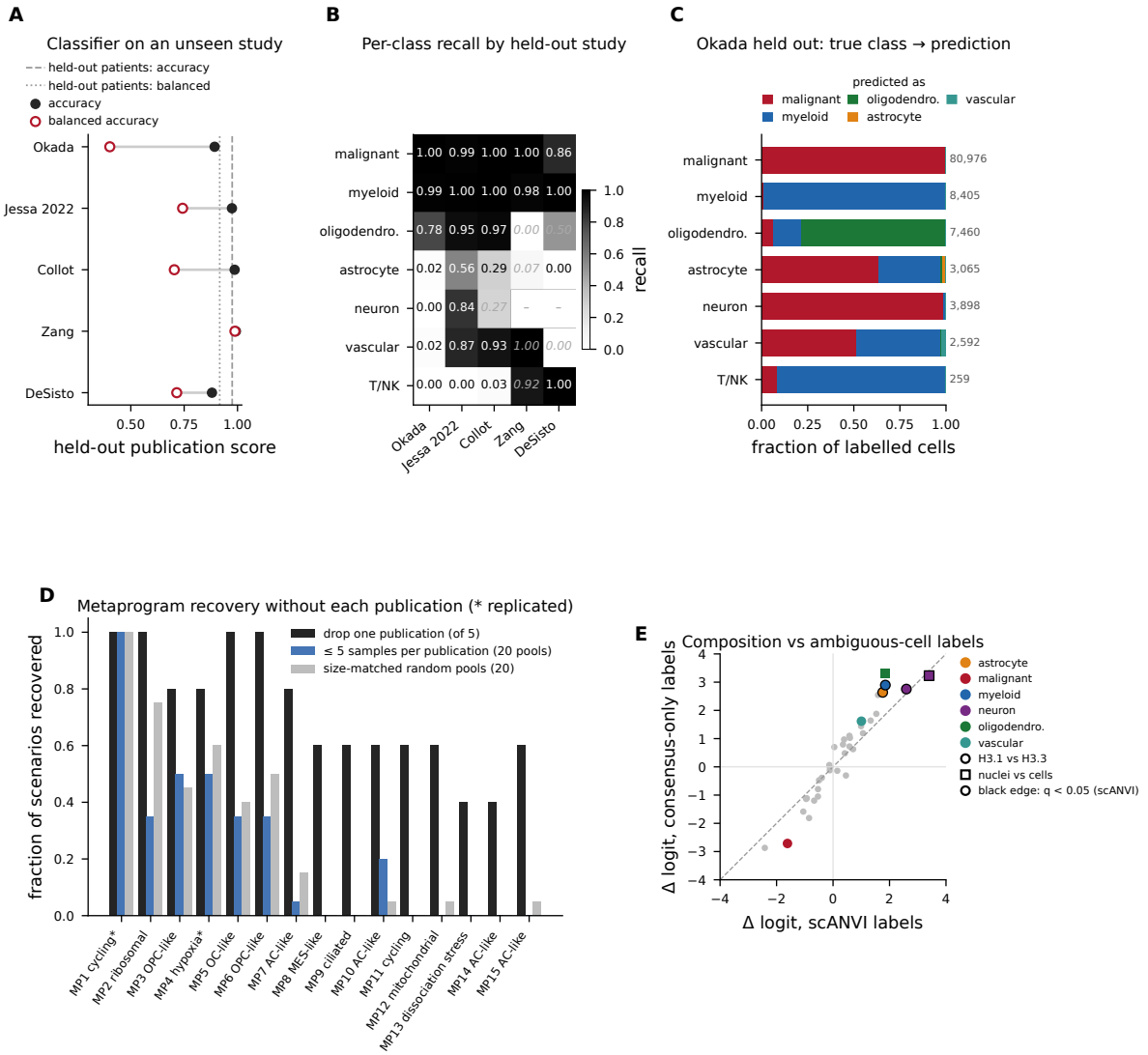


Fig. 6 | Robustness of the reference to removing one study. **a**, Accuracy and balanced accuracy on each held-out publication after retraining scVI and scANVI without it; dashed and dotted lines mark the held-out-patient accuracy and balanced accuracy of the released model. **b**, Per-class recall for each held-out publication; grey italic values are based on fewer than 20 cells. **c**, Predicted labels for each true class of the post-mortem series when that series is held out. **d**, Fraction of scenarios in which each metaprogram is recovered (gene-set Jaccard similarity ≥ 0.5); asterisks mark the two replicated metaprograms. **e**, Composition coefficients under scANVI labels against consensus-only labels; coloured points are significant ($q < 0.05$) under at least one labelling, with a black edge when significant under scANVI labels.

tumour references, and copy-number or marker evidence to support normal-lineage calls in new data.

The co-localisation of MES-like tumour cells with myeloid and vascular cells is consistent with the hypoxia-associated mesenchymal compartment described in glioblastoma[20] and with bone morphogenetic protein (BMP)-driven invasive phenotypes in DMG[25]. The brainstem gradient is consistent with an infiltrative tumour front but rests on a single autopsy. Engineered brainstem organoid models of H3.3K27M DMG[44] provide a means to test these associations experimentally.

Limitations

This study has several limitations. One post-mortem series contributes half of the cells and every autopsy sample, so autopsy and resection cannot be compared. All H3.1 samples are pontine nuclei, and the H3 variant is unreported for 25 of 57 samples. The whole-cell arm of every assay comparison rests on six samples. The reference lacks normal progenitor and ependymal classes, and on unseen studies the classifier often misclassifies rare normal lineages as malignant. Copy-number thresholds were tuned on 22 samples, and 11.2% of consensus-malignant reference cells rely on the aggregate copy-number score without a supporting arm-level event. In 21 of 50 samples the selected NMF rank was the largest tested ($k = 9$), so some samples may contain more programmes than were resolved. The spatial conclusions are based on 12 patients with high between-patient heterogeneity. Finally, two query cohorts were deposited without raw counts.

Methods

Data collection

In October 2026 we screened GEO series (78 candidate series in two searches) and the OpenAlex literature database (393 records) for human DMG data generated by single-cell or single-nucleus RNA sequencing or Visium spatial transcriptomics. Data sets were included if they contained primary human H3 K27-altered midline glioma or normal human brain tissue and provided processed expression matrices; reasons for exclusion are listed in Supplementary Table 1. The reference comprised GEO GSE210567 and GSE210566[16], Zenodo 10.5281/zenodo.17922773[23], GSE231859[27], GSE245674 (sample GSM7846769)[35] and the post-mortem series of Okada et al.[26] (processed matrix from its UCSC Cell Browser instance; raw reads under controlled access at the European Genome-phenome Archive, EGAS50000001447). Projection-only cohorts were GSE102130[15], GSE184357[18], GSE227983 and GSE271936, and two normal brain references obtained from CZ CELLxGENE Discover[45]: the hindbrain, pons and medulla dissections of a first-trimester brain atlas[39], of which 100,000 cells were sampled, and six adult pons dissections[38], of which 100,000 nuclei were sampled. Spatial cohorts were GSE268577[25], GSE194329[24], GSE280990[43] and Zenodo 10.5281/zenodo.17922773. All data were human and previously published, and each study obtained its own ethical approval and informed consent.

Gene harmonisation and quality control

Gene identifiers were mapped to the GENCODE v32 annotation of the 10x Genomics GRCh38-2020-A reference, by Ensembl identifier where available and otherwise by exact symbol or unique alias; ambiguous symbols were removed and duplicated genes were summed. The reference used

the 18,882 genes measured in every sample of every UMI study. Quality control was applied per assay class with Scanpy[46]. Nuclei were retained with at least 300 detected genes and a mitochondrial read fraction of at most 5% (20% for four nucleus samples with cytoplasmic contamination); whole cells with at least 300 genes and at most 20% mitochondrial reads; and Smart-seq2 profiles with at least 1,000 genes and at most 20% mitochondrial reads (5% for Smart-seq2 profiles of frozen nuclei). Cells above the 99.5th percentile of detected genes within their sample were removed. Doublets were scored per sample with Scrublet[47] (expected doublet rate 0.06, 3,000 highly variable genes), and cells with a score above 0.25 were removed from the reference. Samples whose retained cells had a median of fewer than 500 detected genes were flagged as low confidence.

Malignant-cell calling

Counts were normalised to 10,000 per cell and log-transformed, and each cell was scored with the Scanpy gene-set scoring function against marker sets for myeloid cells (*PTPRC*, *CSF1R*, *CX3CR1*, *P2RY12*, *C1QA*, *C1QB*, *CD74*), T and natural killer cells (*CD3D*, *CD3E*, *CD2*, *NKG7*), oligodendrocytes (*MBP*, *PLP1*, *MOG*, *MAG*, *ST18*), vascular cells (*CLDN5*, *FLT1*, *VWF*, *PDGFRB*, *RGS5*), neurons (*RBFOX3*, *SYT1*, *SNAP25*, *GAD1*, *SLC17A7*), astrocytes (*AQP4*, *GFAP*) and the OPC-like (*PDGFRA*, *OLIG1*, *OLIG2*, *SOX10*, *CSPG4*), AC-like (*GFAP*, *AQP4*, *APOE*, *CLU*), MES-like (*CD44*, *VIM*, *CHI3L1*, *ANXA1*) and cycling (*MKI67*, *TOP2A*) tumour states. A cell was called normal by markers if its best normal-lineage score was at least 0.25, exceeded the best competing tumour-state score by at least 0.10 and exceeded the astrocyte score; normal cells with a score of at least 0.50 and at least twice the second-best lineage score served as copy-number references.

Copy-number profiles were computed per sample in the manner of inferCNV, using autosomal genes expressed in at least 10% of cells and ordered by GENCODE v32 coordinates[48]. For each cell, log-normalised expression was compared with the mean of each reference lineage, clipped at ± 3, smoothed as a running mean over 100-gene windows with a step of 10 genes and median-centred, and the lineage with the smallest mean absolute deviation was retained as the baseline. Values below 1.5 standard deviations of the sample were set to zero. Profiles were computed within each chromosome and, for arm-level events, within each chromosome arm; the copy-number score of a cell was its mean absolute smoothed deviation. The reference for each sample combined its own immune cells with up to 300 myeloid and T or natural killer cells pooled from other samples of the same study and assay, and with up to 300 oligodendrocyte, vascular and neuronal cells per lineage pooled from other samples after removal of cells in tumour-grade copy-number clusters; a sample's own non-immune cells were never used as reference. Reference cells with a copy-number score above the reference median plus three median absolute deviations were removed, and samples with fewer than 30 reference cells were not called. Cells were clustered by the Leiden algorithm (resolution 1.0) on principal components of the copy-number profile. A cluster was called malignant if its mean copy-number score exceeded the reference mean by 0.008, if one recurrent arm-level event (gain of 1q, 2 or 7; loss of 10q, 13q, 14q or 17p) reached $z \geq 2$ relative to the reference, or if two events reached $z \geq 1$; clusters containing at least 20% reference cells additionally required both score elevation and an event. The consensus call was malignant or normal when the marker and copy-number calls agreed (and, where available,

the author label agreed), and ambiguous otherwise. The permissive variant additionally called malignant those marker-normal cells that lay in a malignant cluster with a supporting arm-level event and had no contradicting author label. Agreement with author labels was computed per sample.

Integration

Of the shared genes, 3,000 highly variable genes were selected with the Seurat v3 method using sample as batch[49]. scVI[31] was trained on raw counts with sample as the batch variable, study and assay as categorical covariates and mitochondrial fraction as a continuous covariate (30 latent dimensions, two hidden layers, negative binomial likelihood, batch size 1,024, at most 26 epochs with early stopping on the evidence lower bound of a random 10% validation set, patience 5), with random seeds 0, 1 and 2. Integration was scored with scib-metrics[33] on 60,023 labelled cells subsampled by sample, using a 90-nearest-neighbour graph: iLISI, kBET, batch average silhouette width and principal component regression for each batch variable (sample, study and assay) and graph connectivity for batch correction; and cLISI, label silhouette width, normalised mutual information and adjusted Rand index for biological conservation. Each metric was min–max scaled across the evaluated embeddings, and the score $0.4 \times \text{batch} + 0.6 \times \text{biology}$, specified in advance, selected among the scVI seeds; Harmony[36] and uncorrected principal components served as benchmarks. Uniform manifold approximation and projection (UMAP) was used for visualisation only[50].

Label transfer and query mapping

scANVI[32] was initialised from the selected scVI model with the consensus lineage labels (ambiguous cells unlabelled) and trained on all labelled cells for at most 25 epochs. Early stopping (patience 5) monitored the classification loss on a validation set of whole patients, comprising about 10% of labelled cells and containing every class, and the weights of the best epoch were restored. Query cohorts were restricted to the 3,000 reference genes and encoded by the reference model without further training. For the two cohorts deposited as TPM (Filbin et al. and LaBelle et al.), pseudo-counts were derived by rescaling each cell's TPM values so that their sum equalled the number of detected genes multiplied by the median counts per detected gene of a matching count-based cohort (Liu et al. RSEM counts for Smart-seq2 data and Jessa et al. 10x data for 10x data), and rounding; Liu et al. expected counts were rounded. These cohorts were used for projection only. Labels were transferred by the scANVI classifier and by a majority vote of the 15 nearest reference cells in the latent space. Mapping uncertainty was the mean distance of a query cell to its 15 nearest reference cells, expressed as a percentile of the same quantity among 50,000 reference cells[34].

Metaprograms

Malignant cells were those with a consensus call of malignant that passed quality control. For each sample with at least 200 such cells, genes detected in at least three cells were reduced to 2,000 highly variable genes (Seurat v3), normalised to 10,000 counts per cell and log-transformed. Non-negative matrix factorisation (scikit-learn; random initialisation, coordinate descent, 500 iterations) was run with ten random restarts for each rank k from 4 to 9; components were

clustered by average linkage on cosine distance, and the consensus programme was the median of each cluster[40]. The selected rank was the largest k whose clustering had a silhouette of at least 0.90 and whose clusters each contained components from at least 80% of restarts. Each programme was represented by its 50 genes with the highest gene spectra scores. Programmes from all samples were clustered by average linkage on Jaccard distance; at a cut of 0.85, clusters containing at least five programmes from at least three samples were metaprograms, each represented by up to 50 genes present in at least 25% of its member programmes. A metaprogram replicated if its member programmes came from at least three publications and from both cells and nuclei; the two Jessa et al. series counted as one publication. The cut and the rule were specified before inspection of the results. Metaprograms were labelled by the reference gene set with the largest significant overlap (hypergeometric test, Benjamini–Hochberg $q < 0.05$, at least five shared genes), and metaprograms whose 50 genes included at least 15 ribosomal-protein genes, at least 5 mitochondrial genes, or at least 10 genes of a curated immediate-early and heat-shock set were classified as technical.

For the masking control, malignant-cell counts of each Jessa et al. sample were summed into pseudobulk profiles ($\log_2$ counts per million). A gene was assay-biased if it differed by at least two-fold in the same direction in both patients profiled as both cells and nuclei, and differed in that direction between the 4 cell and 23 nucleus samples (Welch t-test, Benjamini–Hochberg $q < 0.05$); mitochondrial and ribosomal-protein genes were added. The 6,013 assay-biased genes were removed from each programme before re-clustering, and the result was compared with ten random masks that removed the same number of genes from each programme.

Per-cell metaprogram families were assigned by scoring each metaprogram with UCell[51] (maximum rank 1,500), standardising scores within sample, averaging the standardised scores of the metaprograms in each family, and taking the family with the highest positive score. Technical metaprograms, and metaprograms whose median UCell score in normal myeloid or vascular cells was at least twice that in malignant cells, were excluded from the families. Cells with a highest family score of at least 1 were used to build the spatial reference.

Microenvironment subclustering

Cells passing quality control and not called malignant were assigned to a compartment (myeloid, T and natural killer, vascular or oligodendroglial) by their consensus lineage label or, for ambiguous cells, by a scANVI label with probability of at least 0.9. Each compartment was clustered by the Leiden algorithm (resolution 0.5) on a 15-nearest-neighbour graph of the scVI latent space. Except in the T and natural killer compartment, subclusters with fewer than 50 cells, or whose mean marker score for another lineage exceeded that for their own, were removed. For the remaining cells, a separate scVI model was trained per compartment on 2,000 highly variable genes after removal of assay-biased, mitochondrial and ribosomal genes (20 latent dimensions, assay class as a covariate, at most 400 epochs with early stopping, patience 20), and cells were re-clustered (Leiden, resolution 0.5). Subclusters were annotated by the reference marker set with the highest mean score and characterised by Wilcoxon rank-sum tests against the other subclusters of the compartment. Assay purity was the fraction of cells in subclusters with less than 10% or more than 90% nuclei.

Composition models

Four composition sets were modelled: lineages of all cells passing quality control (scANVI labels); malignant states (OPC-like, AC-like, OC-like, MES-like and cycling, assigned by the highest within-sample standardised gene-set score); metaprogram families (standardised within assay class); and microenvironment subclusters within each compartment (samples with at least 30 cells). The contrasts were thalamus versus pons, adult versus paediatric, H3.1 versus H3.3, dordaviprone (ONC201) with standard care versus standard care alone, autopsy versus resection, and nuclei versus cells; samples with unknown or other levels were excluded from each contrast. For each sample with at least 200 cells (lineages, states and families), the proportion of each category was transformed to an empirical logit, $\log\{(k + 0.5)/(n - k + 0.5)\}$, where k is the number of cells in the category and n the number in the sample. Each contrast was modelled by ordinary least squares with publication as a fixed effect, adjusting for assay class where it varied within a publication, and with standard errors clustered by patient. Contrasts without variation within any publication were not estimated. The H3-variant contrast was refitted with site as a covariate and within pontine samples. P values were corrected by the Benjamini–Hochberg method[41] within each contrast.

Spatial deconvolution and statistics

In-tissue spots from Space Ranger filtered matrices were used, with genes restricted per cohort to those present in every section and in the reference. Reference signatures for cell2location[42] were estimated from 59,993 cells in 13 classes (seven malignant metaprogram families and six normal lineages; at most 8,000 cells per class, sampled within class by sample) with a negative binomial regression model (250 epochs, sample as batch). Each cohort was deconvolved with an expected 5 cells per spot and a detection alpha of 20, for between 2,000 and 20,000 epochs set by a fixed GPU time budget; the realised number of epochs was recorded. Spot proportions were the 5% posterior quantile of each class divided by their sum over classes. Co-localisation between each of 7 malignant families and 6 normal lineages (42 pairs) was measured as the Spearman correlation of spot proportions per section, Fisher z-transformed, averaged within patient, pooled across patients by DerSimonian–Laird random effects[52], and corrected by the Benjamini–Hochberg method. The ten sections from one autopsy were treated as one patient. Spots with fewer than 500 UMIs were removed in a sensitivity analysis. As a further sensitivity analysis, correlations were recomputed on centred log-ratio-transformed proportions.

Robustness analyses

For each of the five publications, scVI was retrained without it using the settings of the released model (seed 1), and scANVI was trained from it with early stopping on a patient-grouped validation set drawn only from the remaining publications. The held-out publication was then mapped by scArches surgery[34] with default parameter freezing, no weight decay and 100 epochs, with its labels hidden and used only for scoring. Balanced accuracy was the mean recall over classes with at least 20 cells in the held-out publication. Metaprograms were re-clustered with the unchanged rule after removing each publication, in 20 pools of at most five samples per publication, and in 20 random pools of equal size; a primary metaprogram was recovered when a metaprogram in the scenario had a gene-set Jaccard similarity of at least 0.5 with it. Re-

clustering of all samples reproduced the primary result exactly. Composition models were refitted under four labellings of ambiguous cells, after confirming that the original labelling reproduced the published coefficients.

Statistics and reproducibility

Patients were the unit of inference for tumour-level claims. Multiple testing was controlled by the Benjamini–Hochberg method within each family of tests, as described above. All analyses were scripted with fixed random seeds and record checksums of their inputs. Software versions are listed in Supplementary Table 2.

Data availability

The atlas data and reference models are available at Zenodo (doi:10.5281/zenodo.23623335).

Code availability

All analysis code and documentation are available at `https://github.com/hossainlab/dmg_atlas`.

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Author contributions

M.J.H. conceived the study, performed all analyses and wrote the manuscript. I.M. mentored the project.

Competing interests

The authors declare no competing interests.

Use of AI-assisted technologies

The authors used Claude (Anthropic) to assist with analysis code, documentation and drafting. The authors reviewed and edited all content and take full responsibility for it.

Supplementary information

Supplementary Figs. 1–7 and Supplementary Tables 1–2 follow the references; Supplementary Data 1–6 are provided as separate Excel files.

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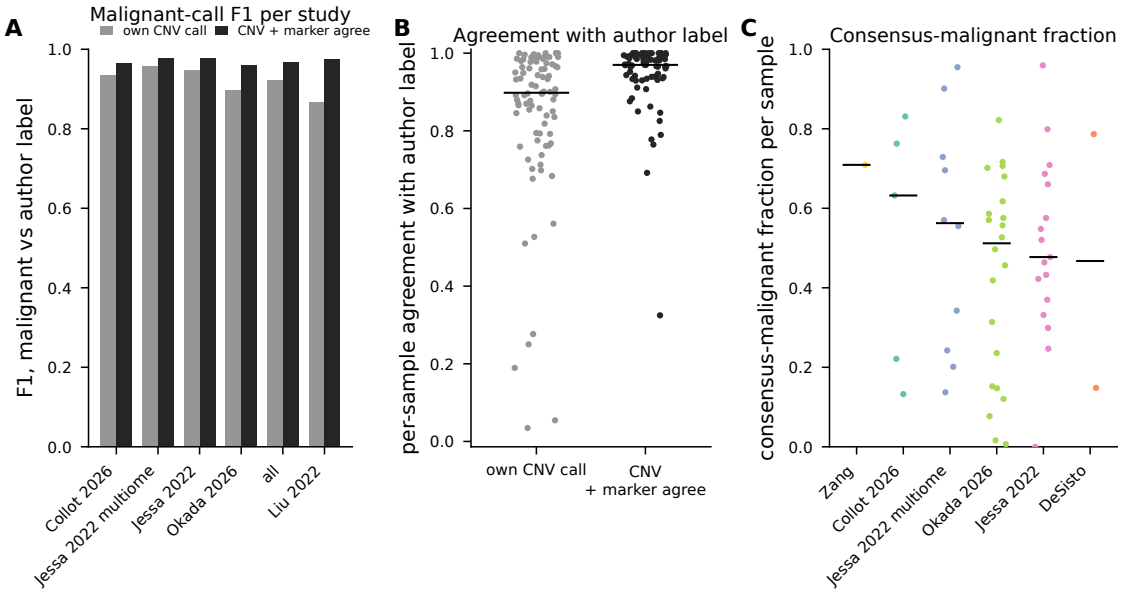
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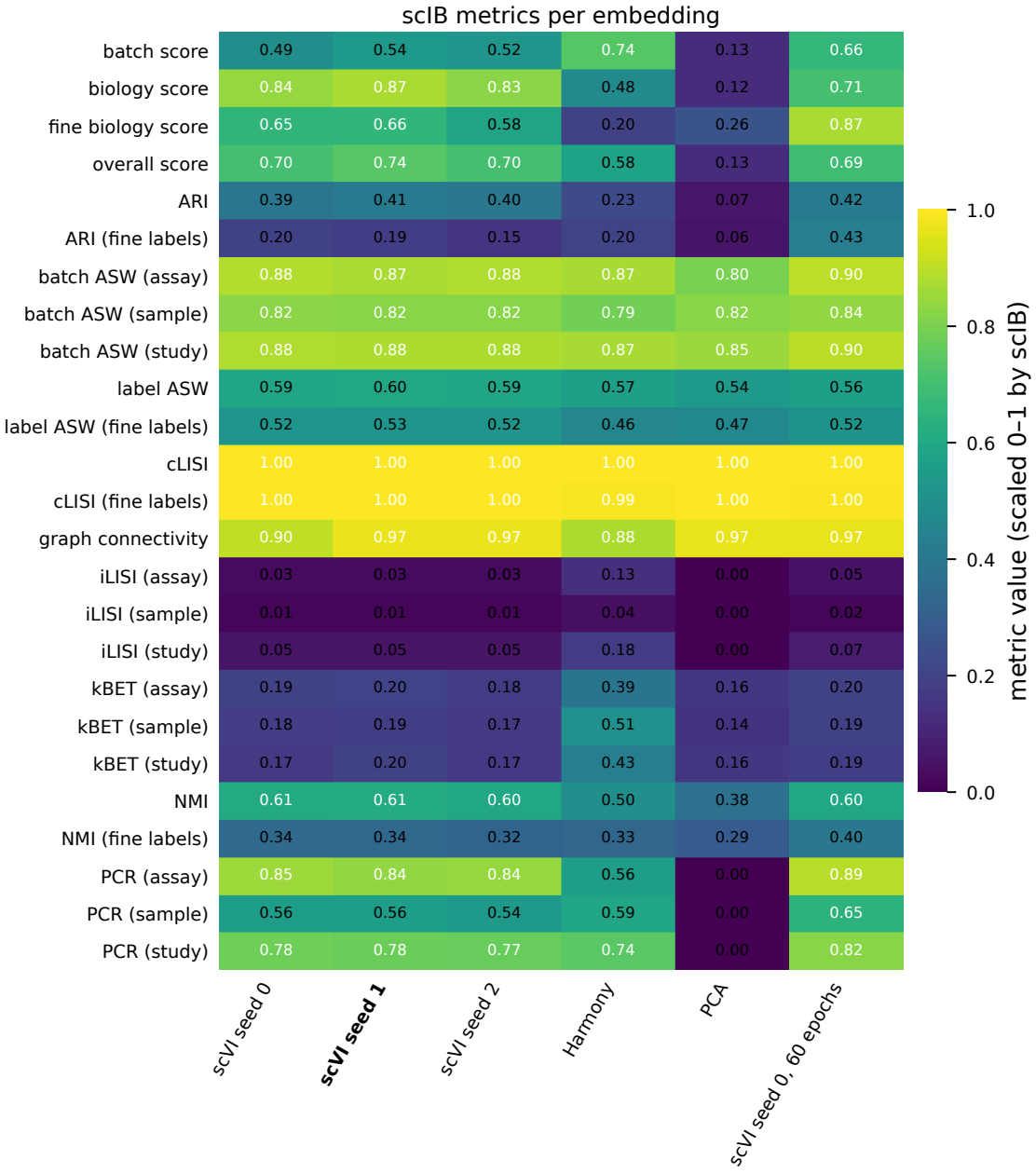
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Supplementary Information

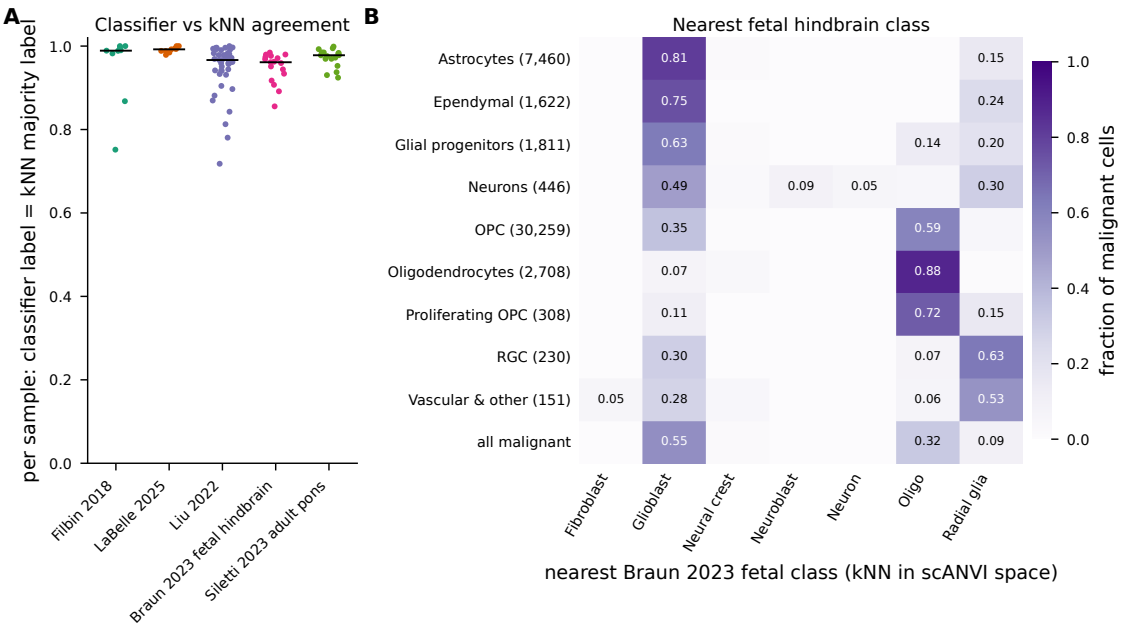
Supplementary Figures



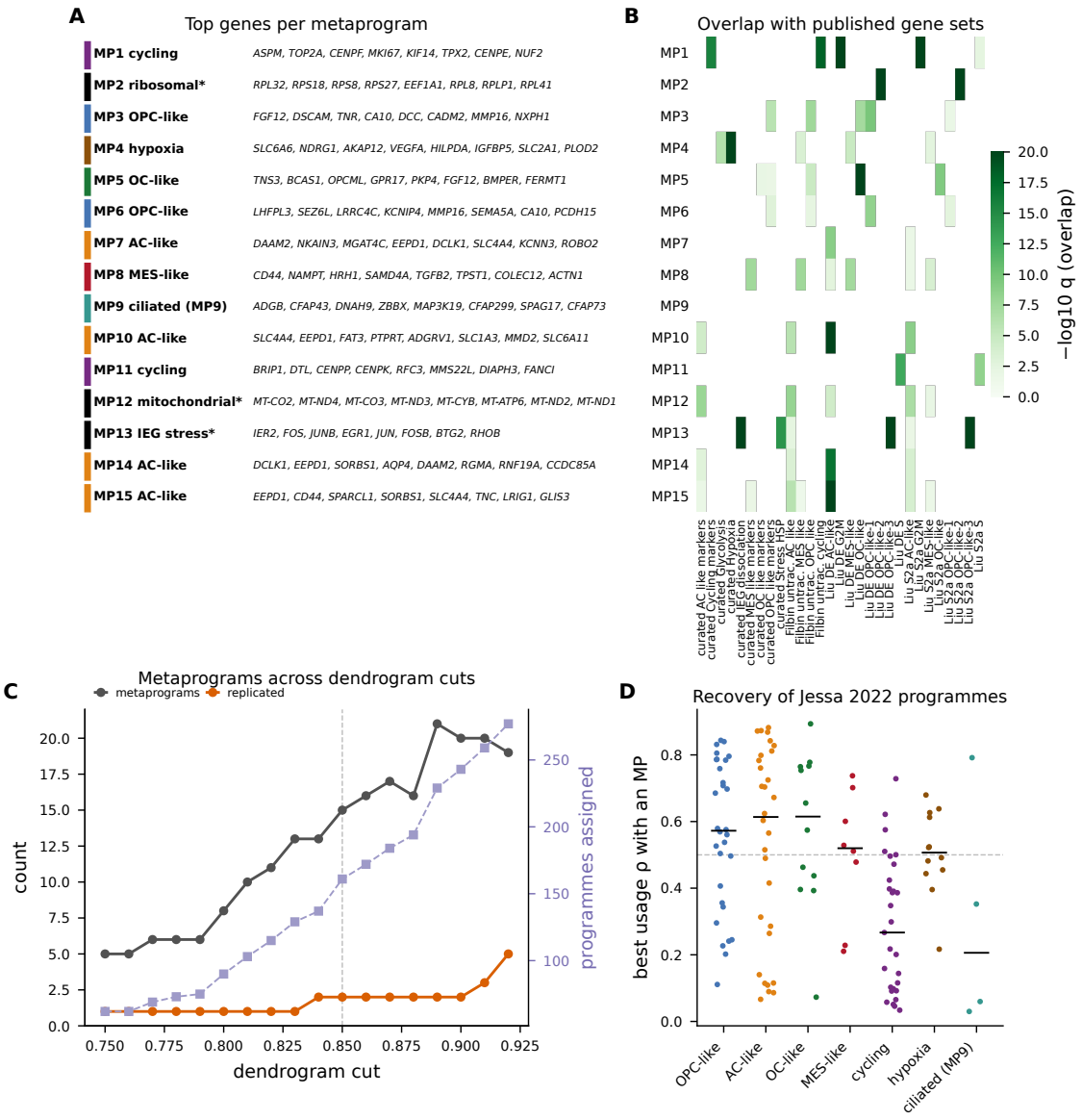
Supplementary Fig. 1 | Malignant-cell calls. **a**, F1 of the author-independent malignant calls against author labels per study: own copy-number call, and cells where copy-number and marker calls agree. **b**, Per-sample agreement with author labels (samples with at least 100 compared cells). **c**, Fraction of consensus-malignant cells per reference sample, by study.



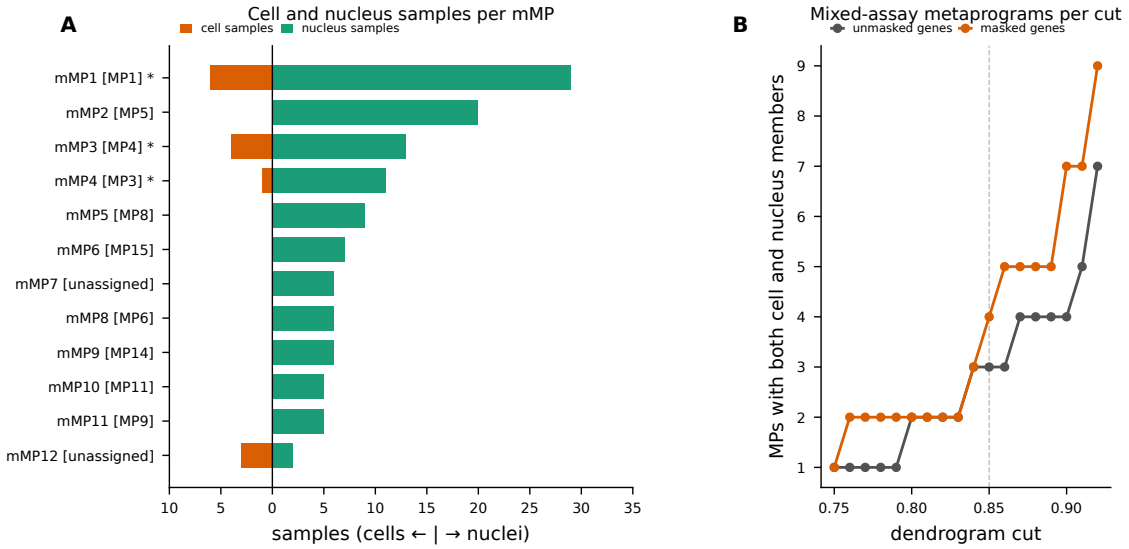
Supplementary Fig. 2 | Integration benchmark. Batch-correction metrics (batch keys sample, study, assay) and biological-conservation metrics (coarse and fine labels) for the six candidate embeddings, with summary scores above. Assay remains the hardest batch key for the chosen model.



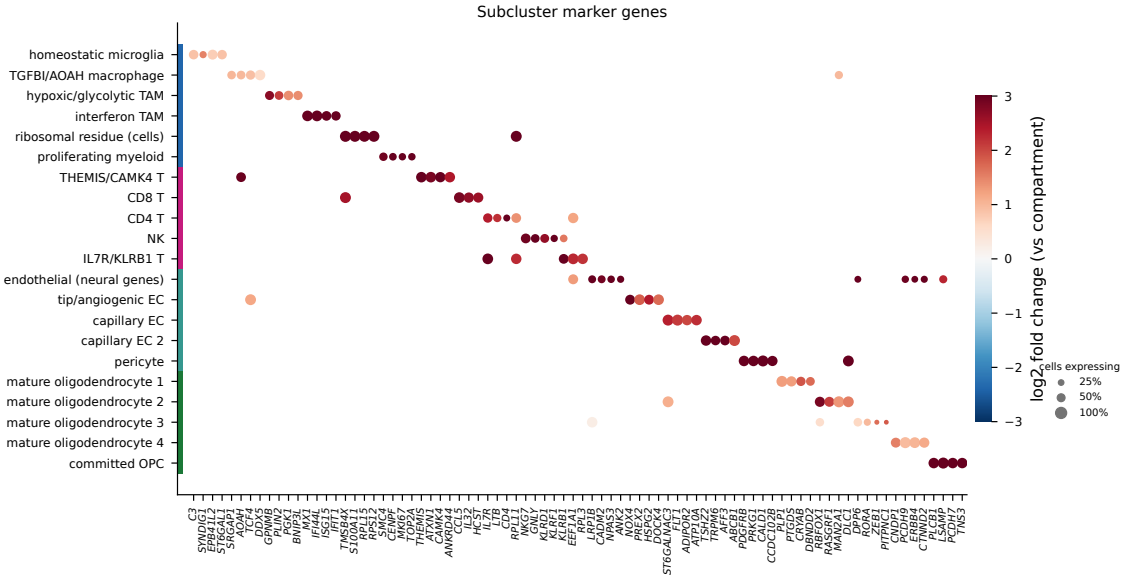
Supplementary Fig. 3 | Mapping confidence and nearest fetal class. **a**, Per-sample fraction of query cells whose classifier label equals the majority label of their 15 nearest reference cells. **b**, Nearest fetal hindbrain class of malignant cells, overall and by author state.



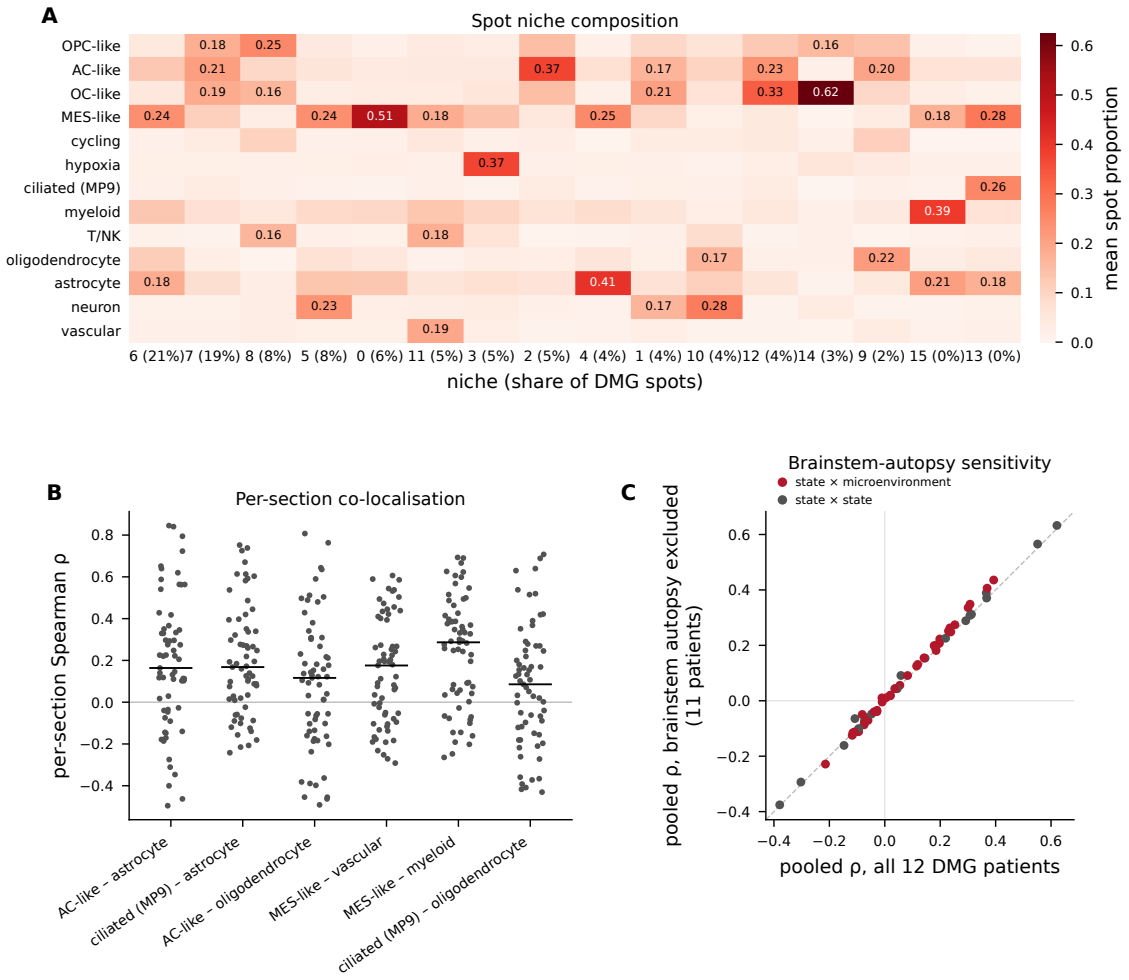
Supplementary Fig. 4 | Metaprogram content and robustness. **a**, Top eight genes of each metaprogram; asterisk, technical programme. **b**, Overlap of metaprogram top-50 genes with published DMG and glioma state gene sets (hypergeometric test, Benjamini–Hochberg q). **c**, Metaprograms, replicated metaprograms and assigned programmes across dendrogram cuts. **d**, Best within-sample correlation of published per-sample programmes with an atlas metaprogram.



Supplementary Fig. 5 | Metaprograms after masking assay-biased genes. **a**, The 12 masked metaprograms at the primary cut, with contributing cell and nucleus samples. **b**, Metaprograms containing both cell- and nucleus-derived programmes across cuts, with all genes and after masking.



Supplementary Fig. 6 | Marker genes of the microenvironment subclusters. Top four genes per subcluster; dot size, fraction of the subcluster's cells expressing the gene; colour, log fold change against the other subclusters of the same compartment.



Supplementary Fig. 7 | Spatial niches and sensitivity analyses. a, Mean class proportions of the 16 spot niches. **b**, Per-section correlations for the six strongest pairs of Fig. 5d, computed with three metrics (proportion, centred log-ratio and abundance); each point is one section and metric, and ten sections come from one autopsy. **c**, Pooled correlations with and without that autopsy.

Supplementary Table 1 | Public data sets screened and their use in this study. Data sets used, with their role, followed by data sets screened but not used, with the reason.

Data set	Accession	Repository	Data type	Use in this study
Collot et al., DMG	10.5281/zenodo.17922773	Zenodo	snRNA-seq	Reference
DeSisto et al., paediatric HGG	GSE231859	GEO	scRNA-seq	Reference
Jessa et al., DMG	GSE210566	GEO	Multiome (RNA)	Reference
Jessa et al., DMG	GSE210567	GEO	scRNA-seq, snRNA-seq	Reference
Okada et al., post-mortem DMG	UCSC Cell Browser (brain-dmg-multiomics); raw data EGAS50000001447	EGA / UCSC Cell Browser	snRNA-seq	Reference
Zang et al., DMG	GSE245674 (GSM7846769)	GEO	scRNA-seq	Reference
Braun et al., first-trimester human brain	CELLxGENE collection 4d8fed08	CZ CELLxGENE	scRNA-seq	Query (normal brain)
Filbin et al., H3K27M glioma	GSE102130	GEO	Smart-seq2 (TPM)	Query (tumour)
LaBelle et al., DMG immune cohort	GSE227983, GSE271936	GEO	Smart-seq2, 10x (TPM)	Query (tumour)
Liu et al., H3-K27M DMG	GSE184357	GEO	Smart-seq2 (TPM)	Query (tumour)
Siletti et al., adult human brain	CELLxGENE collection 283d65eb	CZ CELLxGENE	snRNA-seq	Query (normal brain)
Whole-brainstem DIPG autopsy Visium	GSE280990	GEO	Visium	Spatial
Collot et al., DMG Visium	10.5281/zenodo.17922773	Zenodo	Visium	Spatial
Huchede et al., DMG Visium	GSE268577	GEO	Visium	Spatial
Ren et al., DMG and glioblastoma Visium	GSE194329	GEO	Visium	Spatial
Adult human brain Paired-Tag (Xie et al.)	GSE327101	GEO	Paired-Tag	Not used: normal reference not required
PBTA / OpenPedCan v14	–	OpenPedCan	Bulk RNA-seq	Not used: bulk data, outside scope
ScPCA SCPCP000021	SCPCP000021	ScPCA portal	snRNA-seq	Excluded: failed quality control
Mouse developing pons and forebrain (Jessa et al.)	GSE188625	GEO	scRNA-seq (mouse)	Excluded: non-human
Longitudinal paediatric high-grade glioma	–	dbGaP	snRNA-seq, snATAC-seq	Not accessed: controlled access
DIPG CAR-T autopsy cohort	–	–	snRNA-seq	Not available: data not released
DIPG single-cell cohort (2026)	–	EGA	scRNA-seq	Not available: not deposited
Cell-line and PDX series	GSE212782 and others	GEO	scRNA-seq	Excluded: models, not primary tumour
CSF and PBMC immune series	GSE261235, GSE186802, GSE202174	GEO	scRNA-seq	Excluded: not tumour tissue

Supplementary Table 2 | Software and versions. Main software packages used in the analysis.

Software	Reference	Source
python v3.11.16	Python Software Foundation	python.org
numpy v2.4.6	Harris et al.	PyPI: numpy
scipy v1.17.1	Virtanen et al.	PyPI: scipy
pandas v2.3.3	McKinney	PyPI: pandas
anndata v0.12.19	Virshup et al.	PyPI: anndata
scanpy v1.11.5	Wolf et al.	PyPI: scanpy
scikit-learn v1.7.2	Pedregosa et al.	PyPI: scikit-learn
statsmodels v0.15.0	Seabold and Perktold	PyPI: statsmodels
matplotlib v3.11.2	Hunter	PyPI: matplotlib
infercnvpy v0.6.1	Sturm et al. (inferCNV port)	PyPI: infercnvpy
leidenalg v0.12.0	Traag et al.	PyPI: leidenalg
umap-learn v0.5.12	McInnes et al.	PyPI: umap-learn
scvi-tools v1.4.2	Lopez et al.; Xu et al.	PyPI: scvi-tools
torch v2.14.1	Paszke et al.	PyPI: torch
cell2location v0.1.5	Kleshchevnikov et al.	PyPI: cell2location
pyro-ppl v1.9.1+ab0491a	Bingham et al.	PyPI: pyro-ppl
lifelines v0.30.3	Davidson-Pilon	PyPI: lifelines

Supplementary Data

Supplementary Data 1–6 are provided as separate Excel files.

File	Content
Supplementary Data 1	Per-sample metadata of the reference and query cohorts (site, age group, H3 variant, assay, accession)
Supplementary Data 2	Consensus metaprograms: summary, top-50 genes, member programmes, replication matrix, cut sensitivity
Supplementary Data 3	Malignant-call agreement with author labels, per sample and per study
Supplementary Data 4	Spatial co-localisation for all pairs and metrics, pooled and per section
Supplementary Data 5	Robustness analyses: leave-one-publication-out classifier, metaprogram recovery, composition label sensitivity
Supplementary Data 6	Composition models: all categories and contrasts, estimability, per-sample proportions